esp32c3-node — Highlights

Phase P8; all 6 recorded gates pass.

At a Glance

A tiny WiFi + Bluetooth LE sensor node, in the spirit of the Seeed XIAO ESP32-C3. Product scope (no mode token), 2 layers, meant to go to JLCPCB as-is with economy PCBA.

- ESP32-C3-MINI-1 module; keep its antenna keep-out clear of copper and parts, antenna at a board edge.
- USB-C for power and flashing over the C3's native USB (no USB-UART bridge); BOOT and RESET buttons.
- Runs from a single-cell LiPo (JST-PH 2-pin) with a charger that has power path, so it runs from USB with or without a battery. 3.3 V LDO for everything. Battery voltage sense into an ADC pin.
- A BME280-class temperature/humidity/pressure sensor on I2C, plus a Qwiic/STEMMA QT connector on the same bus.
- One status LED on a GPIO.
- Keep it small, roughly XIAO-sized if an honest layout allows; spare GPIOs on castellated or 0.1 in edge pads.

FIELD	VALUE
part number	PCB-0020-A
board	esp32c3-node
phase	P8 (Verification)
size	26.0 x 38.0 mm
gates	6 of 6 recorded gates pass

The Board

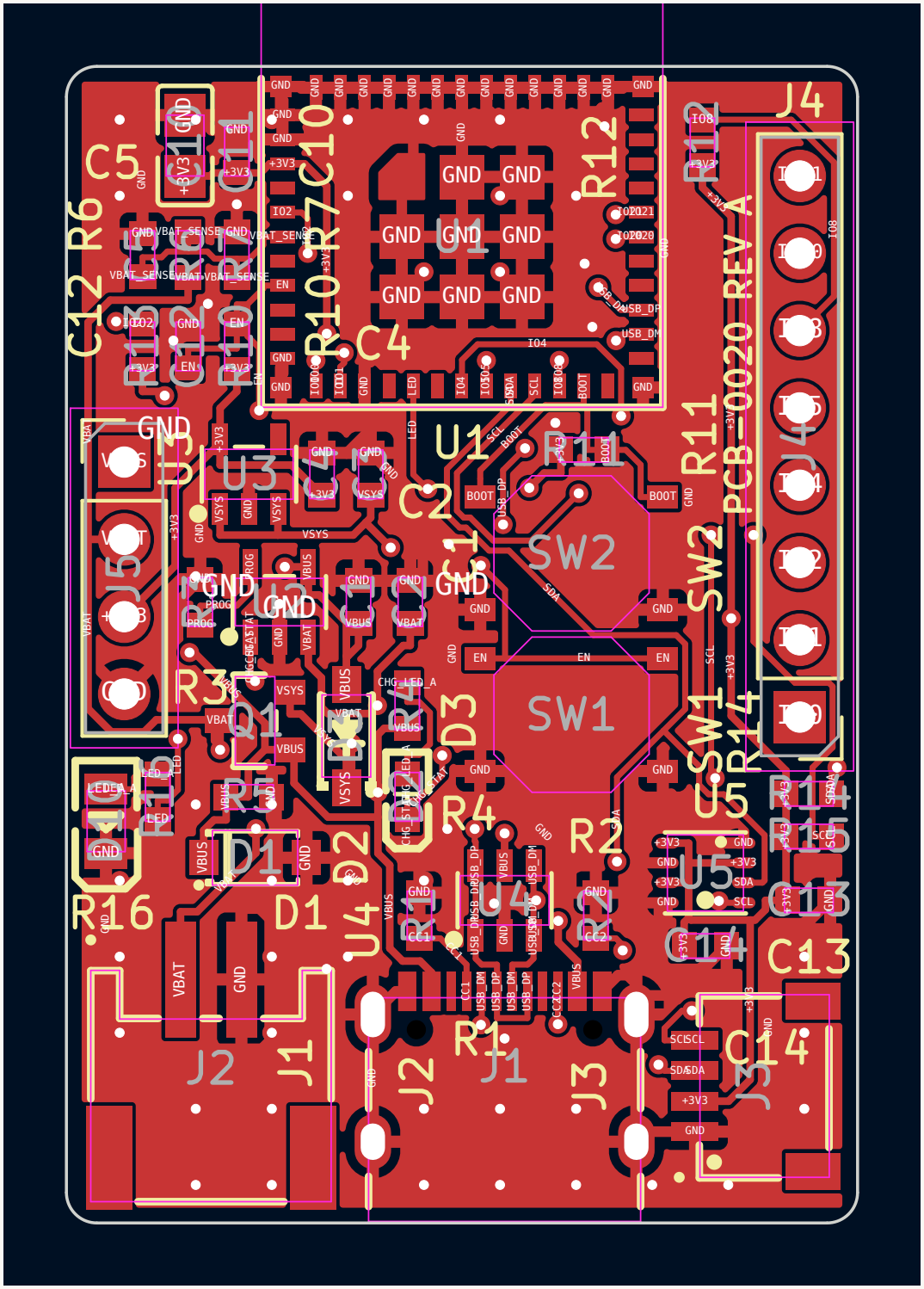


top and bottom, as rendered from the board file

Copper Layers

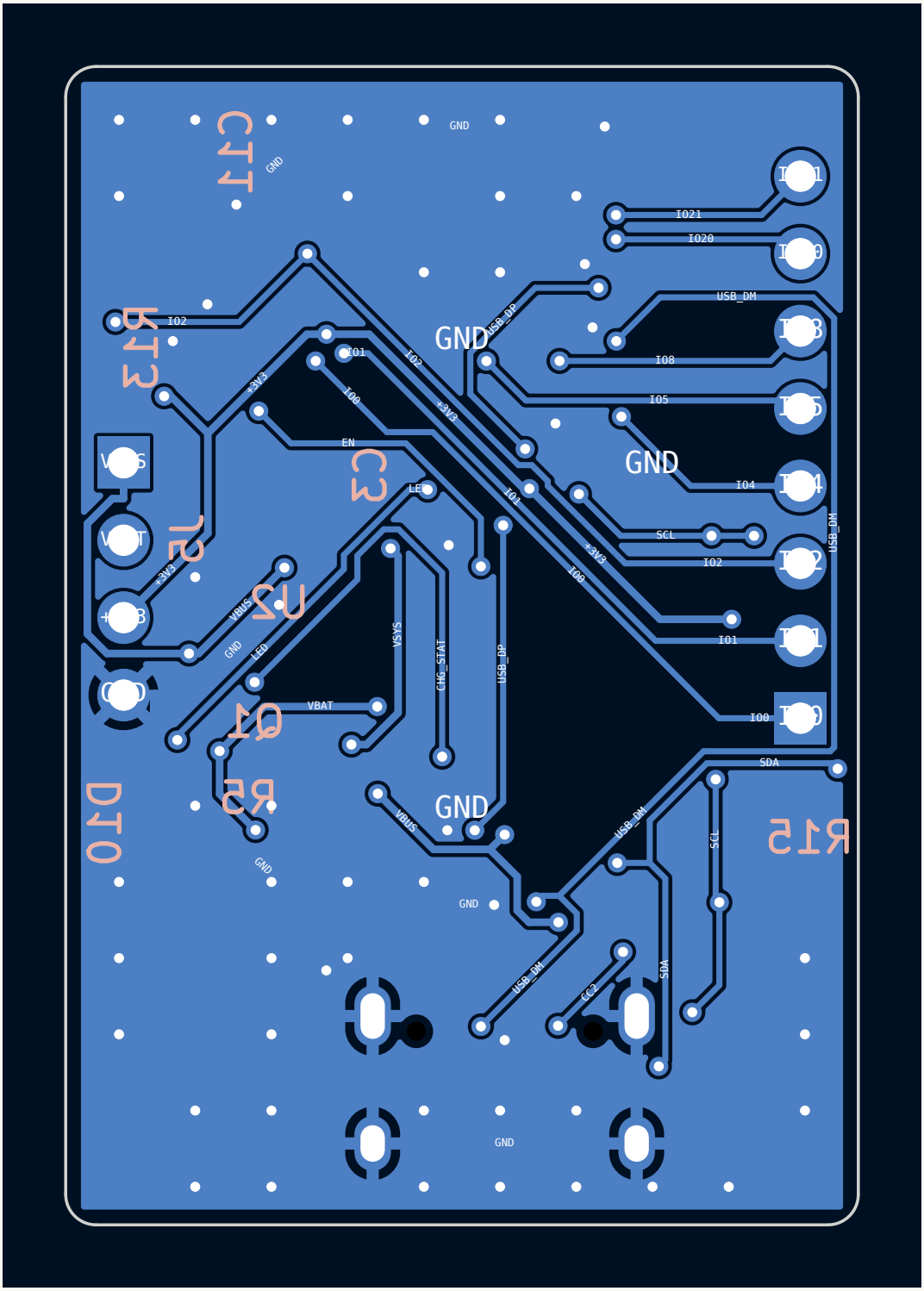
The outer copper layers as KiCad shows them, with every pad, track and zone labelled with its net (zoom in to read the small ones).

F.Cu



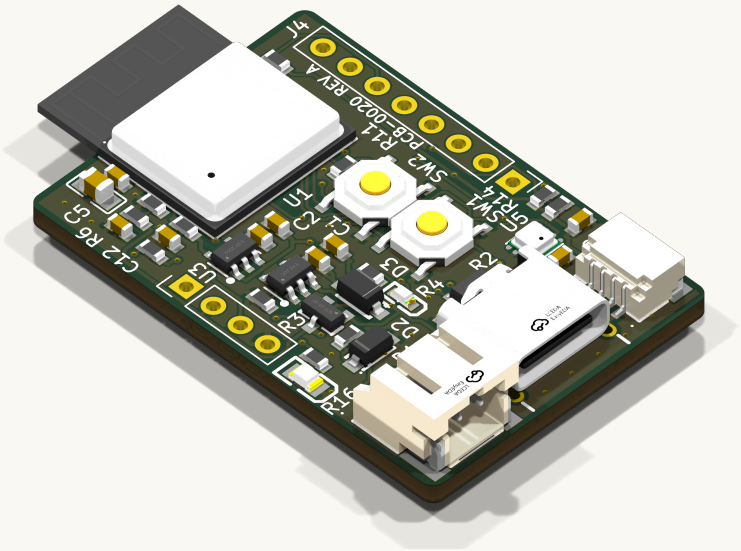
reports/layers/PCB-0020-A_esp32c3-node_F_Cu.pdf

B.Cu



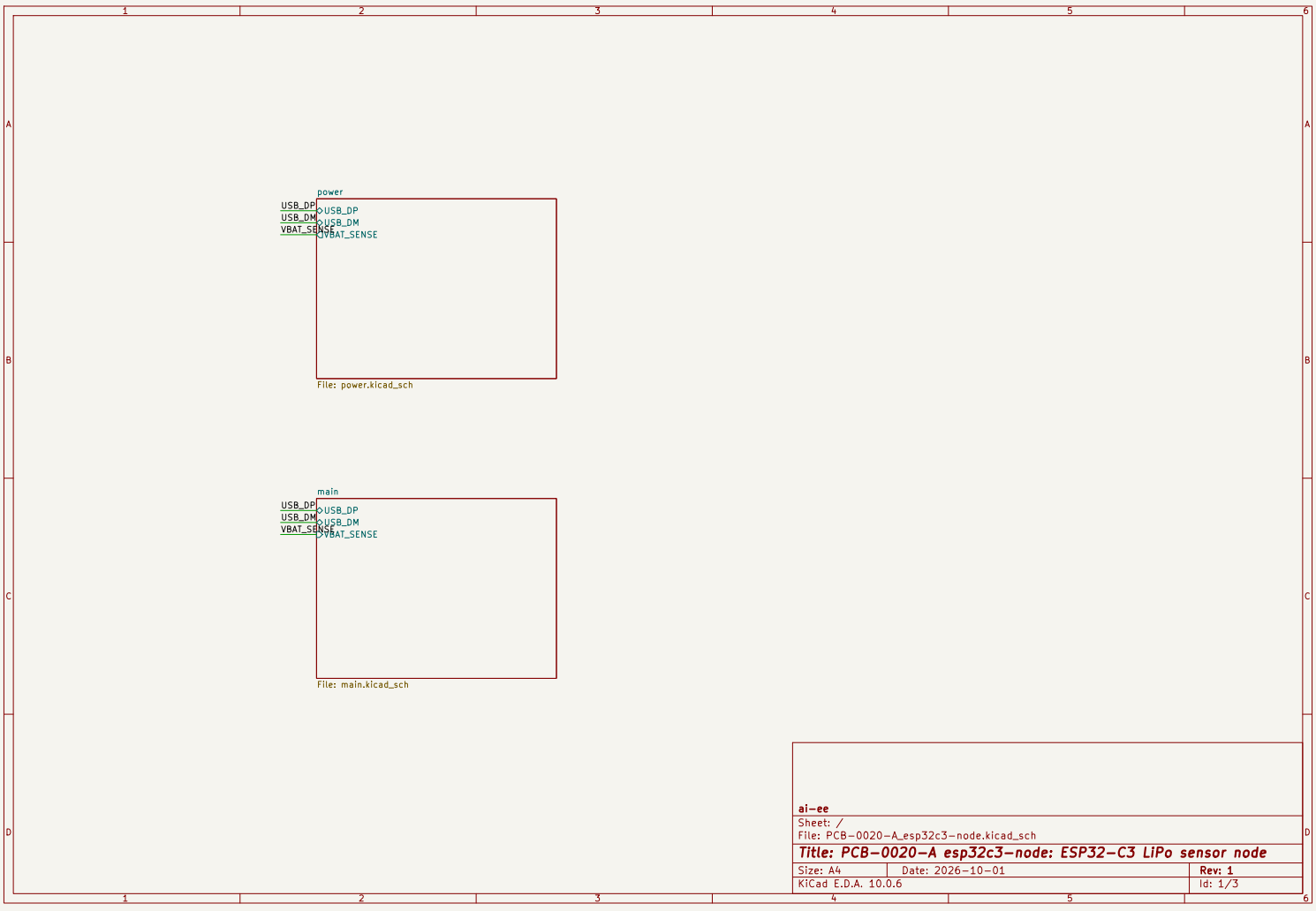
reports/layers/PCB-0020-A_esp32c3-node_B_Cu.pdf

3D view



reports/layers/PCB-0020-A_esp32c3-node_iso.png

What Is On It


power
USB_DP
USB_DM
VBAT_SENSE
USB_DP
USB_DM
VBAT_SENSE
File: power.kicad_sch
main
USB_DP
USB_DM
VBAT_SENSE
USB_DP
USB_DM
VBAT_SENSE
File: main.kicad_sch
ai-ee
Sheet: /
File: PCB-0020-A_esp32c3-node.kicad_sch
Title: PCB-0020-A esp32c3-node: ESP32-C3 LiPo sensor node
Size: A4
Date: 2026-10-01
Rev: 1
KiCad E.D.A. 10.0.6
Id: 1/3

Key Decisions

PH	WHEN	DECISION	WHY
P0	10-01 05:36	Q1b cell class ~400 mAh (300-1000 mAh acceptable); PROVISIONAL (delegate, owner said decide for yourself; re-confirm at H1)	XIAO-sized node class; sizing the charge current to the smallest plausible cell is the conservative choice
P2	10-01 05:48	accepted deviation: VBUS attach charge ~65 uC > USB2.0 50 uC guideline	LDO current limit bounds surge; shown at H1
P3	10-01 06:11	fp_verify U1 (C2838502) exit 1 'pad count 61 != 53' accepted as a checker artifact, not a footprint defect	Footprint has exactly 53 pad numbers; pad 49 (GND EPAD) is a 3x3 array of 1.45 mm squares (one custom pad chamfered for pin-1 orientation, reported as 0.005 mm), total 5.4 mm square as DS Fig 11-1. Edge pads 0.4x0.8 at 0.8 pitch, corners 0.7x0.7, all match....
P3	10-01 06:11	J1 C165948, J2 C160404, J3 C295747 footprints accepted from the JLC/EasyEDA pull without a datasheet extraction	No prior board has extractions for them; spend-bounded run. Pad geometry checked by hand: J2 1.0 pitch 0.6x1.55 + 1.2x1.8 mounts; J3 2.0 pitch 1.0x3.8 + 1.5x3.4 mounts; J1 16-pin USB-C 0.3x1.3 at 0.5 pitch + 4 shell THT. Re-check against the vendor drawings...

How the Run Went

23 recorded decisions from P0 to P3, 2026-10-01T05:36 to 2026-10-01T06:11. The run went back to an earlier phase 0 times.

Checks and Cost

GATE	STATUS	ATTEMPTS	FAILING
erc	pass	5	0
place	pass	5	0
drc_routed	pass	11	0
verify	pass	7	0
dfm	pass	2	0

Model cost of generating this board: $24.98.

The full design doc (esp32c3-node-design-full) has every decision, the renders and the run's history.